Reactivity: units, coefficients, and control

How is reactivity balanced as a reactor moves from cold shutdown to full power?

20 LESSON

Read: FNRP § 9.1–9.4; connections to the four-factor model and control absorption.

Objective: Quantify reactivity and its changes with operating conditions.

1 Measure departure from criticality

The multiplication factor measures gains relative to losses. For an infinite, nonleaking system,

$$k_\infty = \frac{\text{total neutron production rate}}{\text{total neutron absorption rate}}.$$

Reactivity measures departure from criticality:

$$\boxed{\rho = \frac{k-1}{k} = 1 - \frac{1}{k} \quad \longrightarrow \quad k = \frac{1}{1-\rho}.}$$

Use k_∞ for the nonleaking unit cell and k_{eff} for the whole-core example below. Subcritical, critical, and supercritical states have $\rho < 0$, $\rho = 0$, and $\rho > 0$, respectively.

2 Choose units; state the reference

Although k and ρ are dimensionless, reactivity is conventionally assigned in units of $\Delta k/k$. Other numerical representations are

Representation	Numerical value
Fraction ($\Delta k/k$)	ρ
Percent (% $\Delta k/k$)	100ρ
Percent-milli (pcm)	$10^5\rho$
Dollars (\$)	ρ/β
Cents	$100\rho/\beta$

Thus 1 pcm is one part in 10^5, and 1% = 1000 pcm. Here β is the **delayed neutron fraction**: the fraction of fission neutrons emitted later through precursor decay. Since β depends on the fissioning species, dollars are system-specific.

Example: reactivity units

k_∞	ρ	% $\Delta k/k$	pcm
1.01	0.00990	0.990	990
1.30	0.2308	23.08	23080

For $\beta = 0.007$, the first case gives $0.00990/0.007 \simeq 1.4$ \$. Separately, 278 pcm gives

$$\rho_\$ = \frac{0.00278}{0.007} = 0.397\,\$ \simeq 40 \text{ cents}.$$

3 Balance excess reactivity with control

Excess reactivity, ρ_{ex}, is the positive reactivity available without removable control. A reserve is needed to offset changes in the system; with no reserve, fuel consumption would move an initially critical system toward subcriticality.

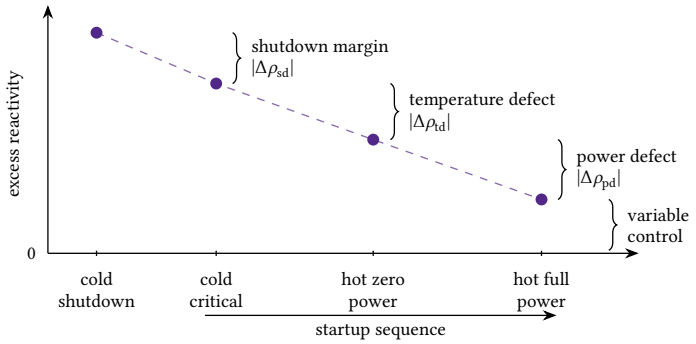
Adjustable control changes absorption to maintain $\rho = 0$. Such control must accommodate the short-term changes in fuel temperature and coolant density and the longer-term changes from xenon buildup and fuel depletion.

A **reactivity worth** is a finite change,

$$\Delta\rho = \rho_2 - \rho_1 = \frac{1}{k_1} - \frac{1}{k_2}.$$

A **reactivity coefficient** is a local derivative, $\alpha_x = \partial\rho/\partial x$, with the other independent parameters fixed. Integral changes connect operating states; coefficients describe local sensitivity.

4 Reactivity Balance During Startup



Cold shutdown is subcritical; the cold-critical, hot-zero-power, and full-power critical states have $\rho = 0$.

Condition	T_M (K)	T_F (K)
Cold shutdown / cold critical	300	300
Hot zero power (HZP)	600	600
Hot full power	600	1200

Shutdown margin, $\Delta\rho_{\text{sd}} < 0$**:** the amount by which the reactor is subcritical with all control rods inserted. At the same cold conditions, the magnitude of the total rod worth is

$$W_{\text{rods}} = \rho_{\text{ex}} + |\Delta\rho_{\text{sd}}|.$$

Temperature defect, $\Delta\rho_{\text{td}} < 0$**:** the change from cold to HZP conditions, before compensating control adjustments. Fuel Doppler broadening and moderator density reduction contribute.

Power defect, $\Delta\rho_{\text{pd}} < 0$**:** the additional change from HZP to full power, again before compensating control. In the stated example, T_F rises from 600 to 1200 K while T_M remains 600 K.

Variable control: rods or soluble boron balance the remaining excess reactivity to restore $\rho = 0$.

5 Modeling Control with the Four-Factor Formula

We can model each state in the startup process using the modified four-factor formula

$$k_\infty = \underbrace{\frac{\nu\bar{\Sigma}_{fT}^F}{\bar{\Sigma}_{aT}^F + \bar{\Sigma}_{aT}^{\text{control}} + \varsigma(V^M/V^F)\bar{\Sigma}_{aT}^M}}_{\eta_T f} \; p\,\varepsilon\,,$$

where F and M denote fuel and moderator. Here, the control term is added directly to the fuel absorption cross section. A similar term will be added in L22 to account for the buildup of fission products that are parasitic neutron absorbers.

Some control (e.g., boric acid) is dissolved in the moderator, in which case the control term would instead add to the moderator absorption cross section. Typically, solutions are sufficiently dilute to ignore impacts on things like $\xi\bar{\Sigma}_s^M$ when evaluating the four factors.